

Buyan Khurelbaatar

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EDUCATION

San Jose State University

San Jose, CA

Bachelor of Science in Computer Science, Mathematics

Expected May 2027

- Relevant Coursework: Data Structures and Algorithms, Object Oriented Design, Operating Systems, Database Management Systems, Distributed Systems

TECHNICAL SKILLS

Languages: Javascript, Typescript, C, Java, C++, Python, SQL

Frameworks, Libraries: React, Spring Boot, Node.js, Next.js, Flask, FastAPI, Redux, GraphQL

Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB), Supabase, Firebase, Pinecone, Snowflake

Cloud DevOps: AWS (EC2, S3, Lambda), Docker, Github Actions (CI/CD), Git, Postman, Jira

EXPERIENCE

SJSU Software and Computer Engineering Society

San Jose, California

Software Engineering Intern

August 2025 - Present

- Led a 3-developer team using Agile and Jira to build the club's website with Next.js, React, and Supabase, deploying to Vercel with a CI/CD pipeline to support 250+ members
- Designed workshops on AI and web development, growing active member participation to 100+ attendees
- Implemented Prometheus based monitoring, alerting, and dashboards to track uptime and basic system metrics for a club registration platform supporting 300+ users

Stanford University Levitt Lab

Stanford, California

Research Assistant

June 2025 – August 2025

- Added major improvements to the PDBCleanV2 python library used for structural biology workflows
- Built chain ID standardization and residue renumbering using sequence alignment for fully consistent datasets
- Collaborated on documentation and tutorial notebooks to streamline onboarding for new users and promote adoption of PDBCleanV2 in structural biology pipelines

Foothill College

Jan 2024 – June 2025

Computer Science Teaching Assistant

Los Altos Hills, California

- Mentored 100+ students in Java and Data Structures, increasing course performance by 20 percent
- Created 30+ coding challenges and led debugging sessions to reinforce algorithms, polymorphism and OOP

PROJECTS

Prep St | *React, FastAPI, Pydantic, OpenAPI, PostgreSQL, Next.js, CI/CD, AWS EC2 - Currently serving 1500+ users*

- Built end-to-end SAT prep platform with Next.js, FastAPI, and Supabase, delivering 147+ React components and 13,000+ lines of Python backend code
- Architected RESTful API with 34 database migrations, 17 services, enabling modular reuse across 50+ endpoints
- Designed end-to-end analytics pipeline tracking 10K+ learning data points per student to predict SAT scores, detect learning plateaus before they occur, and optimize study paths
- Configured multi-environment deployment across Railway (backend), Vercel (frontend), and Modal (serverless compute) with Docker containerization
- Developed personalized AI tutoring system integrating LLM's feedback with intelligent caching, achieving \$0.015 cost per 100-question session

Cowabunga IDE | *Typescript, Electron, React Flow, Pinecone, AWS, Terraform, Microservices, Cloud Infrastructure*

- 2D open world infinite canvas, codebase as flow diagram that lets user build using instructions (vscode fork)
- Engineered AST and data-flow analysis with LLMs to deliver semantic understanding and impact tracing
- Integrated cloud vector database storing embeddings, enabling semantic retrieval with 95 percent precision
- Built agentic tool-calling framework for LLMs to autonomously access, modify user files and folders

Clarity | *React, PostgreSQL, Node.js, Tailwind, Pydantic, FastAPI, Cloud Infrastructure, Docker, CI/CD*

- App that creates a daily time block schedule based on the user's long-term goals and their intention for the day.
- Implemented goal tracking and reflection system enabling users to monitor progress across long-term objectives
- Developed scheduling algorithm integrating task duration, priority, and focus time to auto-balance workload
- Created responsive UI with React, TypeScript, and Tailwind CSS for seamless use across devices